
Distilling GPT-4 Preferences for Summarization: Metric Calibration and DPO Training Under Limited Compute

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Abstract

Supervised TL;DR models often produce summaries that are fluent but misaligned with human preferences, including redundancy and subtle hallucinations. We address this problem by building a preference-aligned summarization system using Direct Preference Optimization (DPO). A Flan-T5 baseline is first fine-tuned on a cleaned TL;DR corpus. Diverse candidate summaries are then generated from multiple encoder-decoder models and scored using ROUGE-L, BERTScore, and an NLI-based factuality metric. A small set of GPT-4 judgments is used to calibrate metric weights via simplex grid search, enabling the construction of a filtered preference dataset combining human-vs-model and model-vs-model comparisons. A DPO model trained on this dataset improves factuality and GPT-4-judged quality over the supervised baseline. Our results also highlight a mismatch between automatic factuality metrics and LLM preferences, underscoring the need for more expressive reward signals for alignment.

1 Introduction

Text summarization models fine-tuned with supervised objectives often exhibit a persistent mismatch between likelihood-based training signals and human preferences[10, 7]. In Reddit TL;DR generation, this misalignment manifests as summaries that are grammatically fluent yet suboptimal in qualities valued by humans, such as factual accuracy, relevance, and conciseness[10]. Since the supervised objective encourages imitation of a single reference summary, the model lacks incentives to discourage redundant, unfocused, or weakly grounded outputs, and cannot learn fine-grained distinctions between better and worse candidate summaries[12, 8].

Preference-based optimization offers a principled alternative by directly shaping the model toward human-valued behaviors[9]. Direct Preference Optimization (DPO) is particularly appealing because it operates on pairwise preference data without requiring explicit reward modeling or reinforcement learning. However, constructing high-quality preference pairs typically depends on costly manual annotation or proprietary LLM evaluations, which limits practical applicability in smaller-scale or resource-constrained settings[6].

This work develops a pipeline for constructing preference data and training a preference-aligned TL;DR summarization model. We begin with a Flan-T5-base supervised baseline and generate diverse

candidate summaries using multiple heterogeneous models and decoding strategies[1]. Candidates are evaluated via ROUGE-L, BERTScore, and an NLI-based factuality estimator, and GPT-4 judgments are distilled into calibrated metric weights through simplex grid search[3, 5]. Using these calibrated scores, we construct a filtered set of human-vs-model and model-vs-model preference pairs with factuality and score-gap constraints. A DPO model is trained on these pairs to encourage concise, relevant, and factually grounded summaries. Subsequent sections detail the pipeline and evaluate its effectiveness against the SFT baseline.

2 Related Work

Early work on text summarization relied on extractive methods and supervised sequence-to-sequence models trained with maximum likelihood objectives. With the introduction of large encoder-decoder models such as BART [4] and PEGASUS [11], abstractive summarization improved substantially, yet remained sensitive to hallucination and often misaligned with human preferences.

To address alignment issues, several lines of research investigate evaluation and preference modeling. ROUGE [5] and BERTScore [12] provide lexical and semantic similarity metrics, while NLI-based factuality estimators such as SummaC [3] and QAFactEval [2] have been shown to correlate with factual consistency. Large language models have also been increasingly used as evaluators for summarization quality.

Preference-based learning has gained prominence as an alternative to likelihood training. Reinforcement learning from human feedback (RLHF) demonstrated the effectiveness of using preference signals for aligning language models [8]. More recently, Direct Preference Optimization (DPO) [9] provides a simplified objective that directly optimizes pairwise preferences without explicit reward modeling, and has been applied across various text generation tasks, including summarization.

This project builds on these directions by combining supervised summarization, automatic factuality estimation, LLM-calibrated metric weighting, and DPO-based alignment to produce preference-consistent Reddit TL;DR summaries.

3 Method

This section describes the full preference-alignment pipeline developed for Reddit TL;DR summarization. The method consists of a supervised initialization stage, large-scale generation of stylistically diverse candidate summaries, metric-based scoring with LLM-calibrated weight estimation, automatic construction of preference pairs, and final optimization using Direct Preference Optimization (DPO). The experimental results are deferred to Section 4.

3.1 Supervised Fine-Tuning

We begin with a supervised model trained to imitate human-written TL;DR summaries. Each Reddit post is transformed into a structured prompt containing the subreddit, title, and a sentence-filtered version of the post body. Formally, for an input post x and gold summary $y = (y_1, \dots, y_T)$, the model is trained to minimize the negative log-likelihood

$$\mathcal{L}_{\text{SFT}}(\theta) = - \sum_{t=1}^T \log \pi_{\theta}(y_t \mid y_{<t}, x),$$

where π_{θ} denotes the conditional sequence distribution defined by the model parameters θ . The resulting Flan-T5 model serves as both a strong baseline and an anchor for subsequent preference optimization.

3.2 Candidate Generation

To expose the preference learner to a broad space of summary behaviors, we construct a large and stylistically diverse candidate pool. Three encoder-decoder models are used, representing distinct summarization tendencies: the supervised fine tuned Flan-T5, which reflects informal and conversational Reddit discourse; DistilBART-CNN, which produces detailed, coverage-oriented summaries; and BART-large-XSum, which yields highly abstractive, compressed paraphrases.

Each model is queried under multiple decoding regimes designed to elicit different error modes and stylistic variations. Deterministic beam search emphasizes coherence, whereas nucleus sampling with moderate temperature introduces controlled diversity. A high-temperature sampling configuration is additionally employed to generate more exploratory outputs, which helps surface hallucinations and edge cases that are informative for preference learning.

3.3 Metric-Based Scoring

To compare candidates without human annotation, each generated summary is evaluated along three complementary dimensions: lexical coverage, semantic similarity, and factual consistency with respect to the original post. These are quantified using ROUGE-L, BERTScore, and an NLI-based factuality estimator, respectively. The factuality score is computed from a DeBERTa-v3 Cross-Encoder model that predicts whether the summary is entailed by the source post; the entailment probability serves as a continuous factuality measure.

Specifically, for a candidate c , we denote these metrics as

$$R(c), \quad B(c), \quad F(c).$$

Although individually informative, these metrics operate on different scales and exhibit varying correlations with human quality judgments, motivating the construction of a unified scoring function.

3.4 Metric Weight Calibration

To obtain a scalar scoring function that aligns with human preferences, we estimate the aggregation weights of ROUGE-L, BERTScore, and factuality using GPT-4 preference labels. Let (c_A, c_B) denote two candidate summaries generated for the same post, and let $z \in \{0, 1\}$ be the GPT-4 judgment, where $z = 1$ indicates that c_A is preferred over c_B . For each pair, we compute the metric differences

$$\Delta = (R(c_A) - R(c_B), B(c_A) - B(c_B), F(c_A) - F(c_B)).$$

We search for non-negative weights $w = (w_R, w_B, w_F)$ on the probability simplex

$$w_R + w_B + w_F = 1, \quad w_i \geq 0,$$

such that the linear score difference

$$D = w^\top \Delta$$

best predicts the GPT-4 preference labels.

Specifically, we discretize the weight space using a step size of $\epsilon = 0.01$ and enumerate all triples (w_R, w_B, w_F) so that each weight lies in $\{0, 0.01, 0.02, \dots, 1\}$, with only those triples retained that satisfy the simplex constraint $w_R + w_B + w_F = 1$. The grid search evaluates every weight configuration against the GPT-4 labels.

For each candidate weight vector w , we predict the preferred summary by $\hat{z}_i = \mathbf{1}[w^\top \Delta_i > 0]$. We then compute the agreement rate

$$\text{Acc}(w) = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[\hat{z}_i = z_i].$$

The final weights $\hat{w}$ are chosen as

$$\hat{w} = \arg \max_w \text{Acc}(w),$$

representing the weight configuration whose induced ranking of summaries most closely matches GPT-4’s preferences.

This grid-based approach provides a simple yet effective non-parametric estimator for metric aggregation.

3.5 Preference Pair Construction

Using the calibrated score $S(c)$, we derive two types of preference pairs. The first contrasts the human TL;DR with the best model-generated candidate whenever the human summary is measurably superior according to the score. The second contrasts the highest- and lowest-scoring model candidates, provided that the better candidate satisfies a factuality threshold and that the score difference exceeds a minimum separation margin.

Formally, for each post x , let

$$c_{\text{best}} = \arg \max_{c \in \mathcal{C}(x)} S(c), \quad c_{\text{worst}} = \arg \min_{c \in \mathcal{C}(x)} S(c).$$

Pairs are constructed only when the score gap

$$S(c_{\text{best}}) - S(c_{\text{worst}}) \geq \delta$$

ensures an unambiguous preference signal. Human-vs-model pairs serve as high-quality alignment anchors, while model-vs-model pairs provide dense discrimination signals across a range of quality levels.

3.6 Direct Preference Optimization

The final model is trained using Direct Preference Optimization (DPO), which directly adjusts the policy to increase the relative likelihood of preferred summaries. Given preference tuples (x, y^c, y^r) where y^c is preferred over y^r , DPO minimizes the loss

$$\mathcal{L}_{\text{DPO}}(\theta) = -\log \sigma \left(\beta \left[\log \pi_{\theta}(y^c | x) - \log \pi_{\theta}(y^r | x) - \log \pi_{\text{ref}}(y^c | x) + \log \pi_{\text{ref}}(y^r | x) \right] \right),$$

where π_{ref} is the frozen SFT model and β controls the strength of the implicit KL regularization.

4 Experiments

This section reports the main empirical results of our system. All methodological details appear in Section 3.

4.1 Supervised Fine-Tuning

We fine-tune a Flan-T5 model on the cleaned Reddit TL;DR corpus using a standard sequence-to-sequence objective. Training is performed until the validation loss stabilizes, yielding a strong supervised baseline. This model serves as the reference policy for preference dataset construction and provides a stable initialization for subsequent DPO training.

4.2 Candidate Generation

Candidate summaries are produced for 15,000 Reddit posts using three encoder-decoder models under the decoding configurations described earlier.

The SFT model is decoded using beam search, moderate-temperature sampling, and high-temperature sampling, producing 45,000 summaries. The two external models are decoded with beam search and high-temperature sampling, yielding 30,000 summaries each. In total, the candidate pool contains 105,000 summaries with substantial stylistic and factual variation that is essential for constructing informative preference pairs.

4.3 Metric Weight Calibration

We randomly sample 400 candidate pairs and obtain GPT-4 preference labels to evaluate how well different metric combinations predict human judgments. Table 1 reports the agreement rates of several weighting schemes. Single metrics achieve 58–63% accuracy. A grid search over the simplex yields the best performance with 67.50% agreement and produces the final weight vector (0.020, 0.780, 0.200) for ROUGE-L, BERTScore, and factuality.

Method	Accuracy	Weights (R,B,F)
ROUGE-only	58.50%	(1, 0, 0)
BERT-only	58.75%	(0, 1, 0)
FACT-only	63.50%	(0, 0, 1)
Grid Search (best)	67.50%	(0.020, 0.780, 0.200)

Table 1: Agreement with GPT-4 preferences on 400 sampled candidate pairs.

4.4 Preference Dataset Construction

Using the calibrated scoring function

$$S(c) = 0.02 \cdot R(c) + 0.78 \cdot B(c) + 0.20 \cdot F(c),$$

we rank all candidate summaries for each post and identify the highest-scoring summary as the *chosen* candidate and the lowest-scoring summary as the *rejected* one. The human-written TL;DR is scored using the same formula. We retain only pairs that satisfy three quality criteria: (i) the factuality of the chosen summary must exceed 0.5; (ii) the score gap between chosen and rejected summaries must be at least 0.05; and (iii) the human TL;DR must achieve a minimum score of 0.3. These filters remove low-quality examples and ensure that each retained pair provides a clear preference signal.

Preference Type	Count	Description
Human > Model	6,569	Human TL;DR clearly outperforms model output
Model > Model	13,227	Best vs. worst model-generated candidates
Total	19,796	High-quality filtered preference pairs

Table 2: Final preference dataset obtained after factuality, score-gap, and human-quality filtering.

4.5 DPO Training

We train three DPO models using the full 19,796 preference pairs, varying only the regularization coefficient $\beta \in \{0.01, 0.05, 0.1\}$. The SFT-tuned Flan-T5 checkpoint serves as both the policy and reference model, and the full model is optimized directly without parameter-efficient techniques. All runs use the same training setup: one epoch, learning rate 5×10^{-6} , gradient accumulation of 8, and FP16 training, a complete list of hyperparameters for all configurations is provided in Appendix B.

4.6 Evaluation

We evaluate the SFT baseline and the three DPO variants using both automatic metrics and large-model preference judgments. All evaluations are conducted on a held-out subset of 500 test posts. For each model, summaries are generated using a standardized beam-search decoding configuration (5 beams, 128-token limit, no-repeat 3-gram constraint) to ensure comparability across conditions.

Automatic Metrics. We report ROUGE-L, BERTScore, and an NLI-based factuality score computed with a DeBERTa-v3 cross-encoder. To highlight how preference optimization modifies the model’s behavior, Table 3 reports both the absolute metrics and their changes relative to the SFT baseline.

Model	ROUGE-L		BERTScore		Factuality	
	Score	Δ	Score	Δ	Score	Δ
SFT Baseline	0.2600	–	0.7889	–	0.4498	–
DPO ($\beta = 0.01$)	0.2518	-0.0082	0.7891	+0.0002	0.5442	+0.0943
DPO ($\beta = 0.05$)	0.2527	-0.0073	0.7901	+0.0012	0.5120	+0.0622
DPO ($\beta = 0.1$)	0.2574	-0.0025	0.7912	+0.0022	0.5020	+0.0521

Table 3: Automatic evaluation metrics for the SFT baseline and DPO models, including score improvements (Δ) relative to the SFT baseline.

LLM-Based Pairwise Evaluation. To measure perceived summary quality, we conduct blind pairwise comparisons using GPT-4 on 100 sampled test posts. For each post, the evaluator is shown the full source text and two candidate summaries. The evaluator must judge which summary better reflects the source content in terms of factual correctness, coverage of key points, and clarity. The exact evaluation prompt is provided in Appendix C. Table 4 reports the win rates of each DPO variant.

Model	Wins	Losses	Ties	Win Rate
DPO ($\beta = 0.01$)	30	56	14	37.0%
DPO ($\beta = 0.05$)	39	36	25	51.5%
DPO ($\beta = 0.1$)	38	34	28	52.0%

Table 4: GPT-4 pairwise evaluation results comparing each DPO model against the SFT baseline. Win rate uses the tie-adjusted formula: $(\text{wins} + 0.5 \times \text{ties}) / \text{total}$.

Analysis of Results. Automatic metrics indicate that all DPO variants improve factuality relative to the SFT baseline, with $\beta = 0.01$ yielding the largest gain. However, this setting also produces the largest drop in ROUGE-L, suggesting increased rewriting and reduced extractive overlap. BERTScore remains stable across configurations, indicating preserved semantic similarity.

In contrast, GPT-4 pairwise evaluation shows a clear preference for the $\beta = 0.05$ and $\beta = 0.1$ models, while the $\beta = 0.01$ model performs substantially worse. This mismatch reveals a limitation of the NLI-based factuality metric: although it rewards reductions in overt contradictions, it fails to penalize subtle hallucinations, which become more frequent when β is too small. Models trained with moderate β values therefore achieve a better balance between stability and preference alignment, resulting in higher GPT-4 win rates.

A qualitative example comparing the SFT baseline and the DPO model is given in Appendix A.

5 Discussion

Our findings highlight both the benefits and limitations of proxy-based preference optimization for TL;DR summarization. While DPO consistently improves semantic alignment and factuality under automatic metrics, these gains do not always translate into higher-quality summaries when judged by large language models. In particular, models trained with very small regularization coefficients (e.g., $\beta = 0.01$) achieve the strongest improvements in factuality metrics yet produce more subtle hallucinations and stylistic drift, leading to markedly lower GPT-4 preference scores. This discrepancy illustrates a key weakness of the NLI-based factuality signal: it detects major contradictions but fails to penalize plausible or stylistically coherent hallucinations.

Moderate values of β (0.05–0.1) yield a more favorable trade-off, preserving the stability of the supervised baseline while still incorporating preference information. These models achieve the highest GPT-4 win rates, suggesting that preference alignment benefits from constrained policy updates that prevent overfitting to the proxy reward model. More broadly, our results show that automatic metrics alone are insufficient to characterize model quality and may misrepresent human or LLM-judged preferences.

Additionally, the reward model is based on a simple linear combination of automatic metrics and cannot fully capture GPT-4’s multidimensional preference structure, limiting its expressiveness.

Overall, this study underscores the need for more expressive reward signals and richer preference data when aligning summarization models. Capturing nuanced hallucinations, stylistic appropriateness, and relevance remains challenging for current automatic metrics, indicating a promising direction for future work on hybrid or learned reward models.

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A Additional Example

This section provides additional qualitative examples comparing the SFT baseline and the DPO model (with $\beta = 0.1$). Each example includes the source post (truncated), the gold TL;DR, and the two model outputs.

Source Post: *[Me (23/F) with my boyfriend (22/M) 6 years, he doesn't remember things he tells me. My boyfriend has odd habit of forgetting important things he says/does. It's been something I've known about him since we met (from forgetting previous actions to forgetting things I witness him telling others), but it really throws a wrench in solving the issues we've been having. Here's an example: He recently invited me to an event related to his job that involves both of our interests. I agreed to go with him. However, a few days before the event I asked him what time we would be leaving that day so I could plan ahead. He completely didn't remember inviting me and told me it was too late since space was limited for transportation to the event. I was kind of hurt, but he kept insisting that for various logical reasons that didn't even make sense that he would have invited me in the first place. This was the second time in the past few months that he forgot about inviting me to go somewhere with him, but it's hard to talk to him about it if he insists on not remembering these things. This happens all the time in various forms and fashions, but again he insists on the fact that if he doesn't remember it, then it didn't happen. He won't remember leaving something out of place when I know I always put that object where it belongs, he will deny that he told his sister he would do something for her even if I was there in the room and could recite the conversation, and sometimes it concerns important things like previous resolutions to previous arguments/disagreements UNLESS another party is there to confirm he did indeed do/say that thing, then and only then will he admit to things he says he doesn't remember doing/saying.]*

Human TL;DR: *My boyfriend forgets he says/does things. This makes communicating issues difficult. How do we deal with a problem if he doesn't remember the problem happened in the first place?*

SFT Output: *Boyfriend doesn't remember things he says/does, and it's hard for me to talk to him about it.*

DPO Output: *Boyfriend doesn't remember things he says/does, and it's hard to talk to him about it because he insists on not remembering things."*

Reason why DPO win: *DPO includes the specific reason why communication is difficult ("because he insists on not remembering things"), which captures the core conflict of the boyfriend denying reality/events based on his poor memory. SFT is more concise but misses this critical nuance of his behavior, making it too generic.*

Table 5: Example comparison between SFT and DPO outputs.

B DPO Training Details

Table 6 reports all hyperparameters used for the three DPO models trained with different values of β .

Hyperparameter	$\beta = 0.01$	$\beta = 0.05$	$\beta = 0.1$
Learning rate	5×10^{-6}	5×10^{-6}	5×10^{-6}
Batch size	4	4	4
Gradient accumulation	8	8	8
Epochs	1	1	1
Max prompt length	1024	1024	1024
Max completion length	128	128	128
Precision	FP16	FP16	FP16

Table 6: Complete hyperparameters for DPO models trained with three different values of β .

C LLM Evaluation Prompt

The ordering of summaries is randomized for every example to ensure unbiased judgment.

```
system_prompt = (
    "You are a strict evaluator of Reddit TL;DR summaries. "
    "Your job is to decide which of two summaries is better for a given post."
)
```



```

user_prompt = f"""
You will be given a Reddit post and two candidate TL;DR summaries (Summary 1 and Summary 2).

[Original Post]
{post}

[Summary 1]
{s1}

[Summary 2]
{s2}

Evaluate the summaries in the following order of importance:
1. Factuality (CRITICAL): A summary must not hallucinate or contradict the post.
2. Coverage and relevance: Prefer the summary that captures the main point or conflict.
3. Conciseness: If both are factual and relevant, prefer the shorter one.
4. Style: Prefer the one that sounds like a natural Reddit TL;DR (informal but clear).

Constraints:
- Avoid ties. Only output "tie" if the two summaries are essentially identical in content and quality.
- If they are very similar, break the tie in favor of the more concise or more natural summary.

Respond STRICTLY in JSON format:

{"winner": "1" | "2" | "tie", "reason": "short explanation"}
```